

Convexity and Convex MPC

Constrained LQR and Receding Horizon Control

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February 21, 2026

- Controllability - Dynamic Programming - Riccati Recursion - Value Function Interpretation

Today:

- Convexity Background - Convex Optimization - Convex Model Predictive Control

Why Convexity Matters

LQR is powerful.

But real systems have:

- Actuator limits - Safety constraints - State bounds

Constraints break Riccati closed-form solution.

However:

If problem is convex $\rightarrow$ we can solve QP online reliably.

A set $\mathcal{C}$ is convex if:

$$x, y \in \mathcal{C} \Rightarrow \theta x + (1 - \theta)y \in \mathcal{C}$$

for all $\theta \in [0, 1]$.

Geometric meaning:

Line between any two points stays inside set.

Examples of Convex Sets

- Linear subspace: $Ax = b$ - Half-space: $a^T x \leq b$ - Polytope: $Ax \leq b$ - Ellipsoid: $x^T P x \leq 1$, $P \succ 0$ - Cones (e.g. second-order cone)
- These appear constantly in control constraints.

A function f is convex if:

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

Equivalent:

Epigraph of f is convex.

Examples:

- Linear functions
- Quadratic with $Q \succeq 0$
- Norms

Convex Optimization Problems

General form:

$$\min f(x)$$

$$\text{s.t. } x \in \mathcal{C}$$

If:

- f convex - $\mathcal{C}$ convex

Then:

Any local minimum is global.

Standard Convex Problems

- Linear Program (LP) - Quadratic Program (QP) -
Quadratically-Constrained QP (QCQP) - Second-Order Cone Program (SOCP)
- LQR with constraints becomes a QP.

Think of MPC as:
Constrained LQR.
We solve:

$$\min_{\{u_k\}} \sum_{k=0}^{N-1} x_k^T Q x_k + u_k^T R u_k + x_N^T P x_N$$

$$x_{k+1} = A x_k + B u_k$$

$$u_{min} \leq u_k \leq u_{max}$$

Receding Horizon Principle

At each time step:

1. Solve finite-horizon QP 2. Apply first control input 3. Shift horizon 4.

Repeat

If no constraints active:

MPC = LQR (for any horizon length).

Role of Terminal Cost

Terminal cost:

$$x_N^T P x_N$$

Approximates infinite-horizon value function.

Better approximation:

- Shorter horizon needed - Better performance

Longer horizon:

- Less reliance on terminal cost.

Computational Advantage

Convex MPC:

- No spurious local minima - Fast Newton convergence - Predictable solve time

This enables:

Real-time embedded control.

Example: Planar Quadrotor

Nonlinear dynamics:

$$m\ddot{x} = -(u_1 + u_2) \sin \theta$$

$$m\ddot{z} = (u_1 + u_2) \cos \theta - mg$$

$$I\ddot{\theta} = \ell(u_2 - u_1)$$

Linearize around hover.

Obtain linear model:

$$x_{k+1} = Ax_k + Bu_k$$

Apply convex MPC with thrust limits.

Convexity guarantees:

- Reliable optimization - Global solutions - Fast convergence

Convex MPC:

Constrained extension of LQR.

Foundation for modern robotics and autonomous systems.